

Vol 10 Chapter 10 — Calculus of Variations and Path Integrals

Keeper (author pass — deep math/physics revision)

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Chapter 10 — Calculus of Variations and Path Integrals

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Level 1 — one sentence

The calculus of variations finds functions extremizing functionals via the Euler-Lagrange equations, providing the foundation for Lagrangian mechanics (Vol 8 Ch 3), classical field theory, and the Feynman path integral — which BST identifies as the substrate's many-Koons-tick coherent sum over K-type intermediate states (Vol 5 Ch 5).

Level 2 — graduate-physicist precision

10.1 Variational principle

For functional $S[\phi] = \int L(\phi, \phi', x) dx$, stationarity $\delta S / \delta \phi = 0$ gives:

$$\frac{d}{dx} \frac{\partial L}{\partial \phi'} - \frac{\partial L}{\partial \phi} = 0$$

The Euler-Lagrange equation. Field-theoretic generalization for $\phi(x^\mu)$:

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} - \frac{\partial \mathcal{L}}{\partial \phi} = 0$$

10.2 Lagrangian mechanics + field theory

Lagrangian mechanics (Vol 8 Ch 3): $L = T - V$, action $S = \int L dt$, equations of motion are Euler-Lagrange.

Classical field theory: $\mathcal{L}$ Lagrangian density, $S = \int \mathcal{L} d^4x$. Euler-Lagrange gives field equations (Maxwell, Klein-Gordon, Dirac, Einstein, Yang-Mills).

10.3 Path integrals

Feynman 1948: quantum amplitude as functional integral:

$$\langle x_f | e^{-i\hat{H}T/\hbar} | x_i \rangle = \int \mathcal{D}\phi(t) e^{iS[\phi]/\hbar}$$

Sum over all paths weighted by $e^{iS/\hbar}$. Classical path is the stationary phase point as $\hbar \rightarrow 0$.

10.4 Saddle-point / stationary phase

For $\hbar$ small: paths off the classical-stationary path have wildly oscillating phase that destructively interferes. Only paths within $\Delta S \sim \hbar$ of the stationary path survive.

This gives WKB approximation and the semiclassical limit (Vol 5 Ch 5 + Vol 8 Ch 3 connection).

10.5 Substrate many-tick reading

Vol 5 Ch 5 Section 5.4: in BST, the path integral is the substrate's many-Koons-tick coherent sum:

$$\langle x_f | e^{-i\hat{H}T/\hbar} | x_i \rangle = \prod_n \langle x_n | e^{-i\hat{H}t_K/\hbar} | x_{n-1} \rangle$$

with $N = T/t_K$ ticks. Continuum limit ($t_K \rightarrow 0$) gives Feynman path integral.

The substrate “tries all paths” in a literal computational sense — Zone 2 evolution evolves all K-type basis elements per tick; only ones whose accumulated phase doesn't destructively interfere survive.

10.6 K-audit anchors

- Vol 5 Ch 5: path integral from substrate
- Vol 8 Ch 3: Lagrangian mechanics
- DCCP: discrete-frame substrate interpretation

Level 3 — 5th-grader accessibility

Calculus of variations: find the function that extremizes some integral. Out come Euler-Lagrange equations. Feynman path integral: quantum amplitudes = sum over all paths $\times e^{iS/\hbar}$. As $\hbar \rightarrow 0$, only the classical-stationary path survives (saddle point). In BST: path integral = substrate's literal many-Koons-tick sum over K-type intermediate states. The classical path is what the substrate “decides on” through constructive interference.

What comes next

Chapter 11 develops asymptotic analysis.

Where to look this up

- Gelfand-Fomin, *Calculus of Variations*
- Feynman-Hibbs, *Quantum Mechanics and Path Integrals*
- BST: Vol 5 Ch 5; Vol 8 Ch 3